Comparison of Pentium Family Processors

1 Architectural Comparison

Feature	Pentium	Pentium II	Pentium III	Pentium 4
Introduction Year	1993	1997	1999	2000
Architecture	P5	P6	Enhanced P6	NetBurst
Word Size	32-bit	32-bit	32-bit	32-bit (with 64-bit extension later)
Clock Speed	60–300 MHz	233–450 MHz	Up to 1 GHz	1.3–3.8 GHz
L1 Cache	16 KB	32 KB (16I + 16D)	32 KB	8–16 KB
L2 Cache	External (on motherboard)	On cartridge (off-die)	On-chip (advanced versions)	On-chip
L2 Cache Speed	Slower than CPU	Half CPU speed	Full CPU speed (newer)	Full CPU speed
Bus Type	Front-Side Bus	Front-Side Bus	Front-Side Bus	Quad-pumped FSB
Bus Speed	66 MHz	66–100 MHz	100–133 MHz	400–533 MHz (effective)
Packaging	PGA Socket	Cartridge (Slot 1)	PGA Socket 370	PGA / LGA
Instruction Set Extensions	MMX	MMX	MMX, SSE	MMX, SSE, SSE2
Parallelism	Superscalar	Superscalar	Superscalar	Superscalar + Hyper-Threading
Power Efficiency	Moderate	Improved	Better	Lower (high power consumption)
Typical Use	Desktop PCs	Desktop / Workstation	Desktop / Server	Desktop / High-performance systems

2 Key Observations (Exam-Oriented)

- Pentium introduced superscalar execution and MMX instructions.
- Pentium II moved L2 cache closer to the CPU using cartridge packaging.
- Pentium III integrated L2 cache on-chip, improving performance significantly.

- Pentium 4 emphasized very high clock speeds using a quad-pumped bus and later introduced Hyper-Threading.
- Performance evolution shifted from frequency scaling to architectural and parallel execution improvements.